Embedded Control for Robotics

Lecture # 13 Discrete time control System

Book: Embedded Digital Control with Microcontrollers

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Overview Today's Lecture

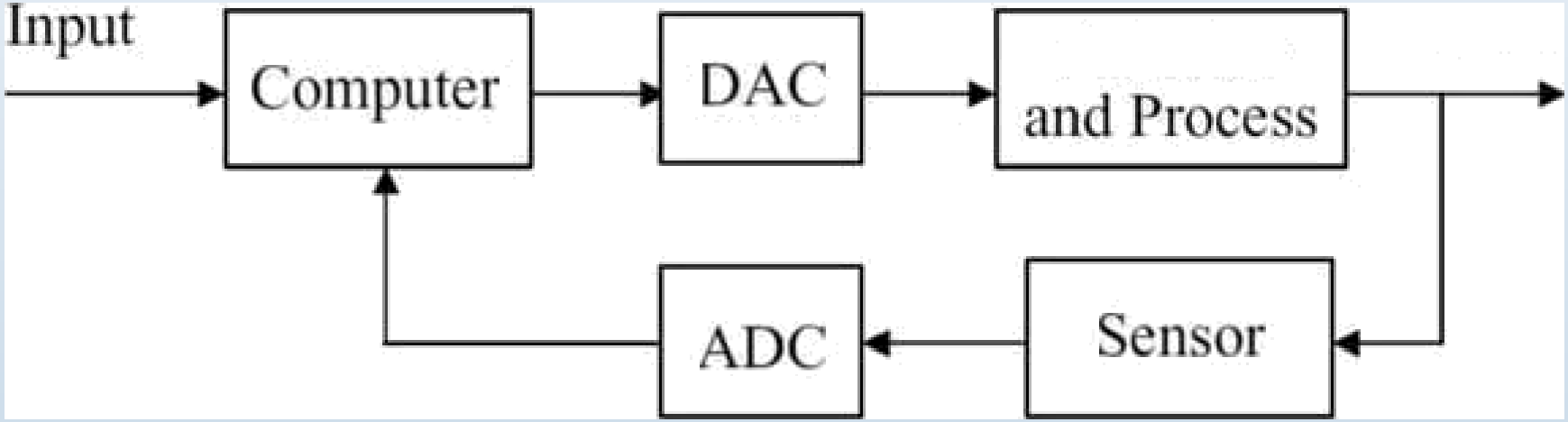
- Introduction
- Difference Equations
- Review of Z-Transform
- Inverse Z-transform
- Relations between s-plane and z-plane
- Solution of difference Equations

Introduction

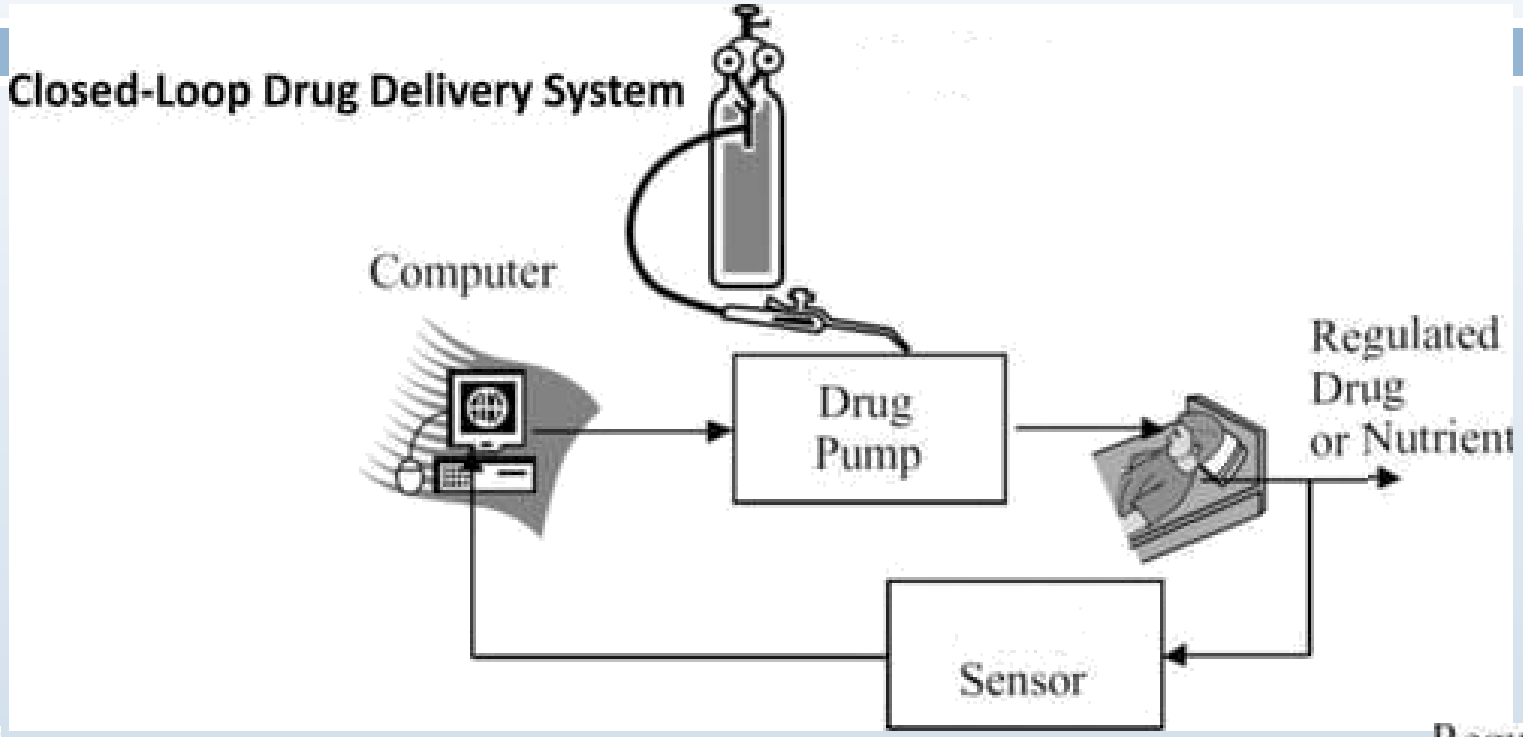
Digital control offers the following distinct advantages over analog control:

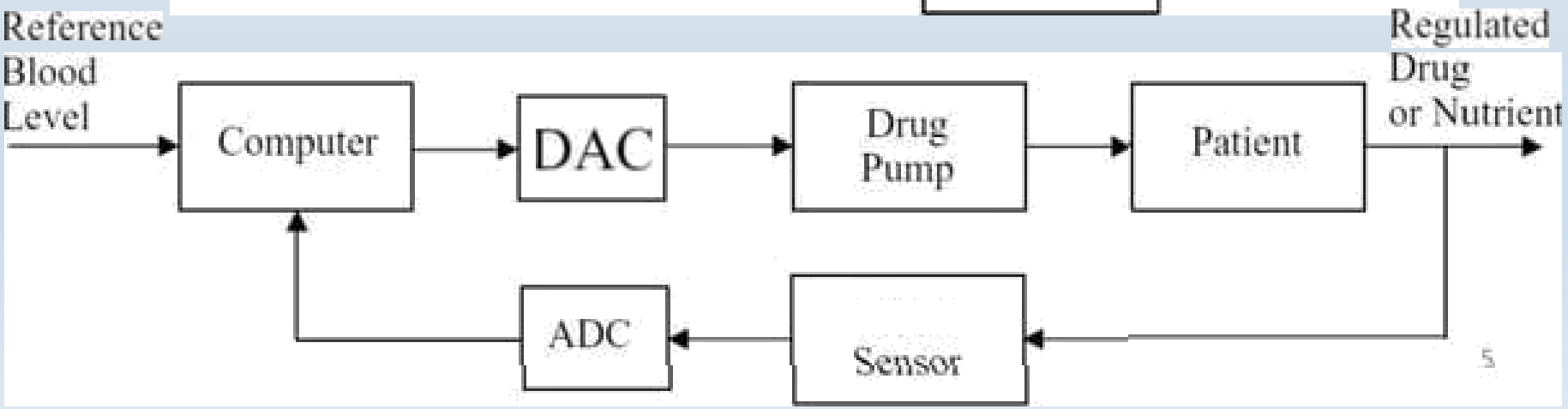
- **Accuracy:** Digital signals are more accurate than their analogue counterparts.
- **Implementation Errors:** Implementation errors are negligible in digital control.
- **Flexibility:** Modification of a digital controller is possible without complete replacement.
- **Speed:** Digital computers may yield superior performance at very fast speeds
- **Cost:** Digital controllers are more economical than analogue controllers.

Digital Control Scheme



Examples of Digital control Systems

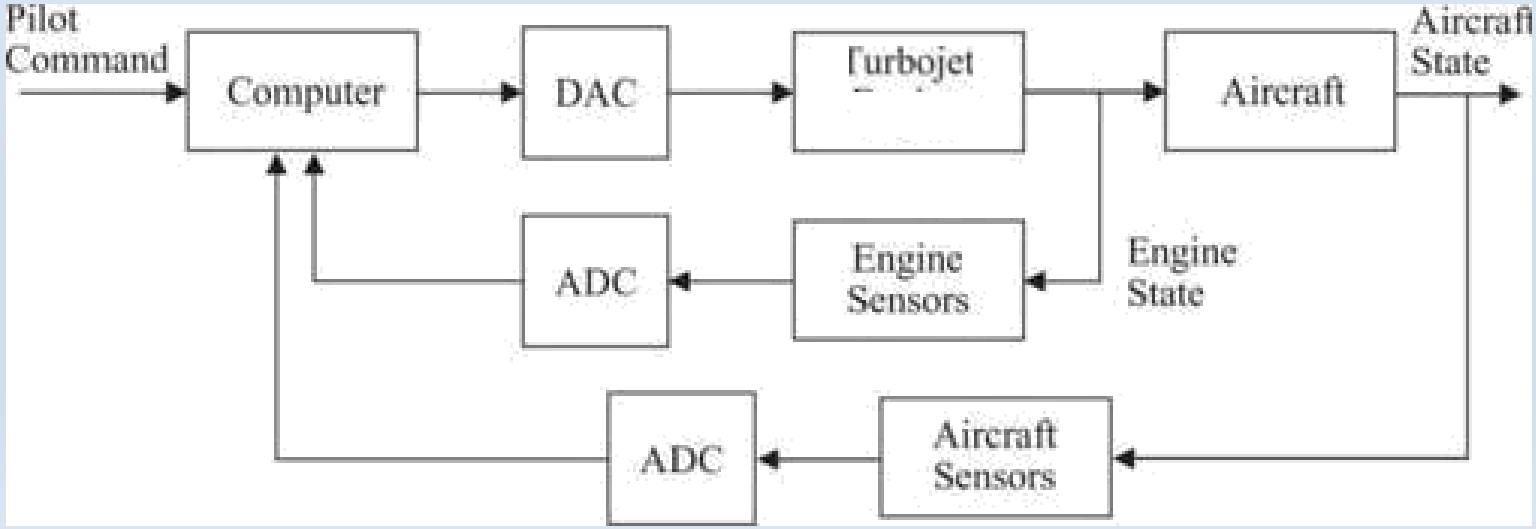




Examples of Digital control Systems

Aircraft Turbojet Engine





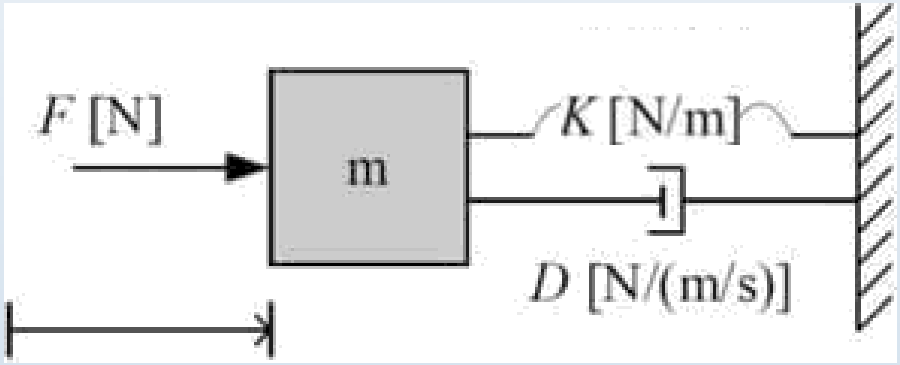
Difference Equation vs Differential Equation

- A **difference equation** expresses the change in some variable as a result of a finite change in another variable.

- A **differential equation** expresses the change in some variable as a result of an infinitesimal change in another variable.

Differential Equation

- Following figure shows a mass-spring-damper-system. Where y is position, F is applied force D is damping constant and K is spring constant.



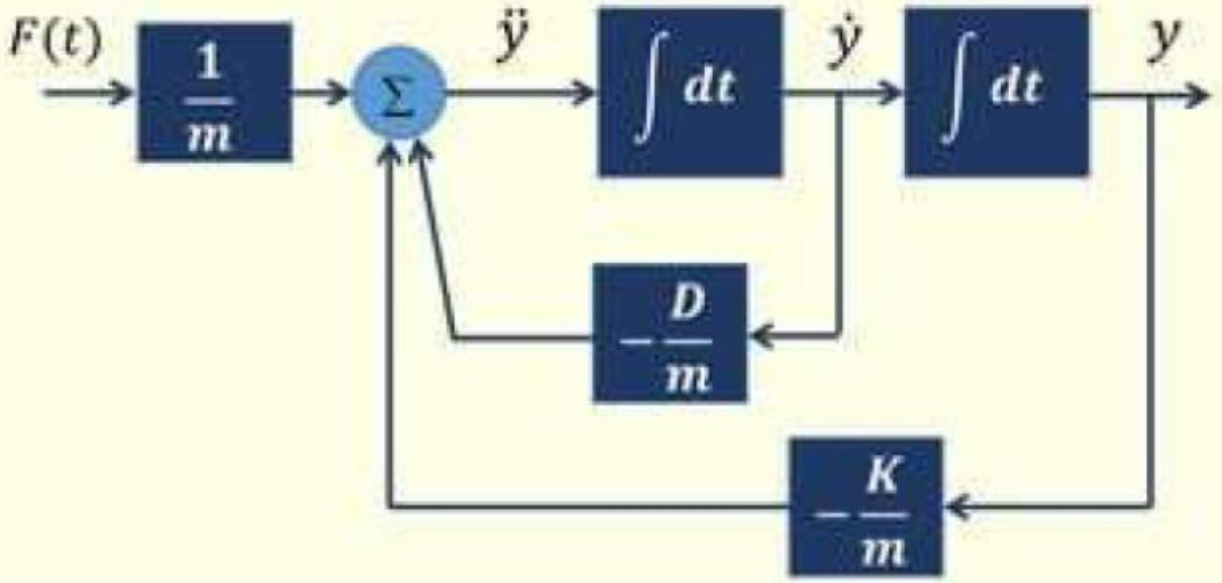
$$F(t) = m\ddot{y}(t) + D\dot{y}(t) + Ky(t)$$

Rearranging above equation in following form

$$\ddot{y}(t) = \frac{1}{m}F(t) - \frac{D}{m}\dot{y}(t)\frac{K}{m}y(t)$$

$$\ddot{y}(t) = \frac{1}{m}F(t) - \frac{D}{m}\dot{y}(t) - \frac{K}{m}y(t)$$

- Rearranging above equation in following form



Difference Equations

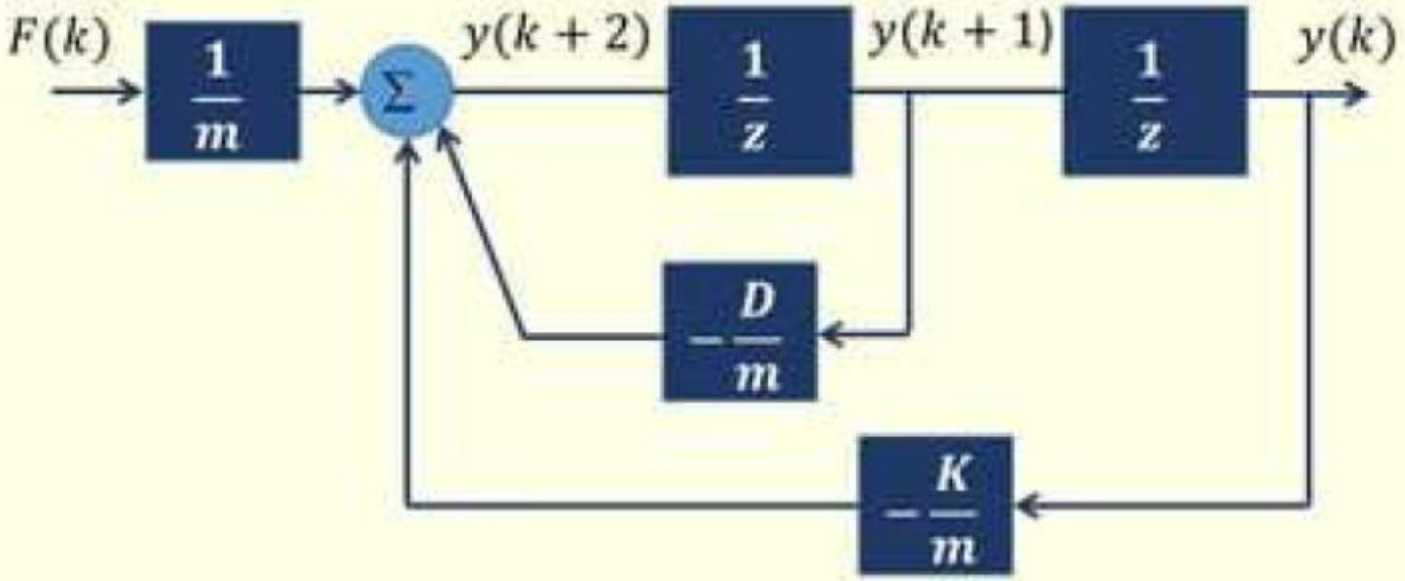
- Difference equations arise in problems where the independent variable, usually time, is assumed to have a discrete set of possible values.

$$y(k+n) + a_{n-1}y(k+n-1) + \cdots + a_1 y(k+1) + a_0 y(k)$$
$$= b_n u(k+n) + b_{n-1}u(k+n-1) + \cdots + b_1 u(k+1) + b_0 u(k)$$

Where coefficients a_{n-1}, a_{n-2},... and b_n, b_{n-1},... are constant.

$u(k)$ is forcing function

$$y(k+2) = \frac{1}{m}F(k) - \frac{D}{m}y(k+1) - \frac{K}{m}y(k)$$



Exp: Difference Equations

- For each of the following difference equations, determine the
  (a) order of the equation
  (b) linear,
  (c) time invariant
  (d) Homogeneous

1. $y(k+2)+0.8y(k+1)+0.07y(k)=u(k)$
2. $y(k+4)+\sin(0.4k)y(k+1)+0.3y(k)=0$
3. $y(k+1)=-0.1y^2(k)$

Exp: 1

1. $y(k+2) + 0.8y(k+1) + 0.07y(k) = u(k)$

Solution:

a) The equation is second order.

b All terms enter the equation linearly

c) All the terms of the equation have constant coefficients. Therefore, the equation is LTI

d) A forcing function appears in the equation, so it is nonhomogeneous.

Exp: 2

2. $y(k+4) + \sin(0.4k)y(k+1) + 0.3y(k) = 0$

Solution

a) The equation is 4th order.
b) All terms are linear
c) The second coefficient is time dependent
d) There is no forcing function therefore the equation is homogeneous.

Exp: 3

3. $y(k+1) = -0.1y^2(k)$

Solution:

a) The equation is 1st order.
b) Nonlinear
c) Time invariant
d) Homogeneous

Z-Transform

- A mathematical tool to convert the difference equations in time domain into the algebraic equations in z-domain
- It is used to describe signals and components in discrete-time control systems.

$$Z[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

$x(n)$	Discrete-time signal — a sequence of samples (for example, values of a signal at times n = 0, 1, 2, ...).
$X(z)$	The Z-transform of $x(n)$ — a function of the complex variable z.
z	A complex variable $z = re^{j\omega}$, where r is magnitude and ω is angle (frequency).
$\sum_{n=-\infty}^{\infty}$	Infinite summation over all integer values of n (from negative infinity to positive infinity).

Z-Transform

- The z-transform is an important tool in the analysis and design of discrete-time systems.
- z-transforms provide a convenient approach for solving LTI Difference equations
- It converts LTI difference equations to algebraic equations and convolution to multiplication.
- It plays a role similar to that served by Laplace transforms in continuous-time problems.

Z-Transform

Given the causal sequence $\{u_0, u_1, u_2, \ldots, u_k\}$, its z-transform is defined as

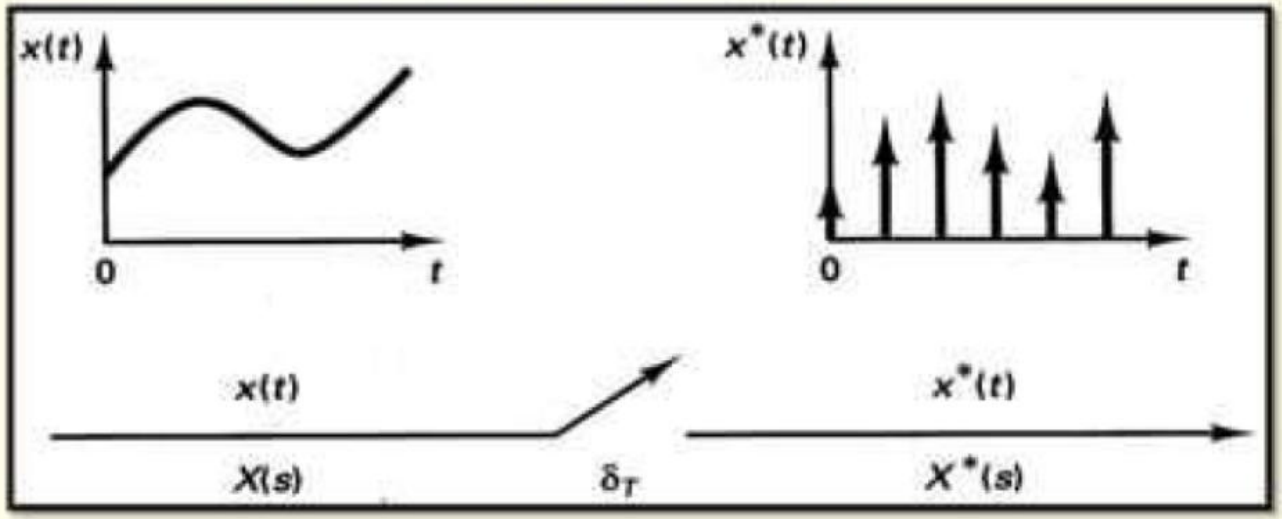
$$U(z) = u_o + u_1 z^{-1} + u_2 z^{-2} + u_k z^{-k}$$

$$U(z) = \sum_{k=0}^{\infty} u_k z^{-k}$$

The variable z^{-1} in the above equation can be regarded as a time delay operator.

Relation between Laplace and Z-Transform

- Given the sampled impulse train of a signal



$$x^*(t) = x(0)\delta(t) + x(T)\delta(t-T) + \cdots + x(kT)\delta(t-kT) + \cdots$$

$$x^*(t) = \sum_{k=0}^{\infty} x(kT)\delta(t-kT)$$

Continue…

Laplace Transform

$$X^*(S) = x(0) + x(T)e^{-sT} + x(2T)e^{-s2T} + \cdots + x(kT)e^{-skT} + \cdots$$

$$X^*(S) = \sum_{k=0}^{\infty} x(kT)\, e^{-skT} = \sum_{k=0}^{\infty} x(kT)\, (e^{sT})^{-k} \qquad \rightarrow (1)$$

- The z-transform is

$$X(z) = \sum_{k=0}^{\infty} x(kT)\, z^{-k} \qquad \rightarrow (2)$$

- Comparing (1) and (2) yields

$$z = e^{sT}$$ where T is the sample period

Mapping between s-plane to z-plane

$$z = e^{sT}$$

Where $s = \sigma + j\omega$.

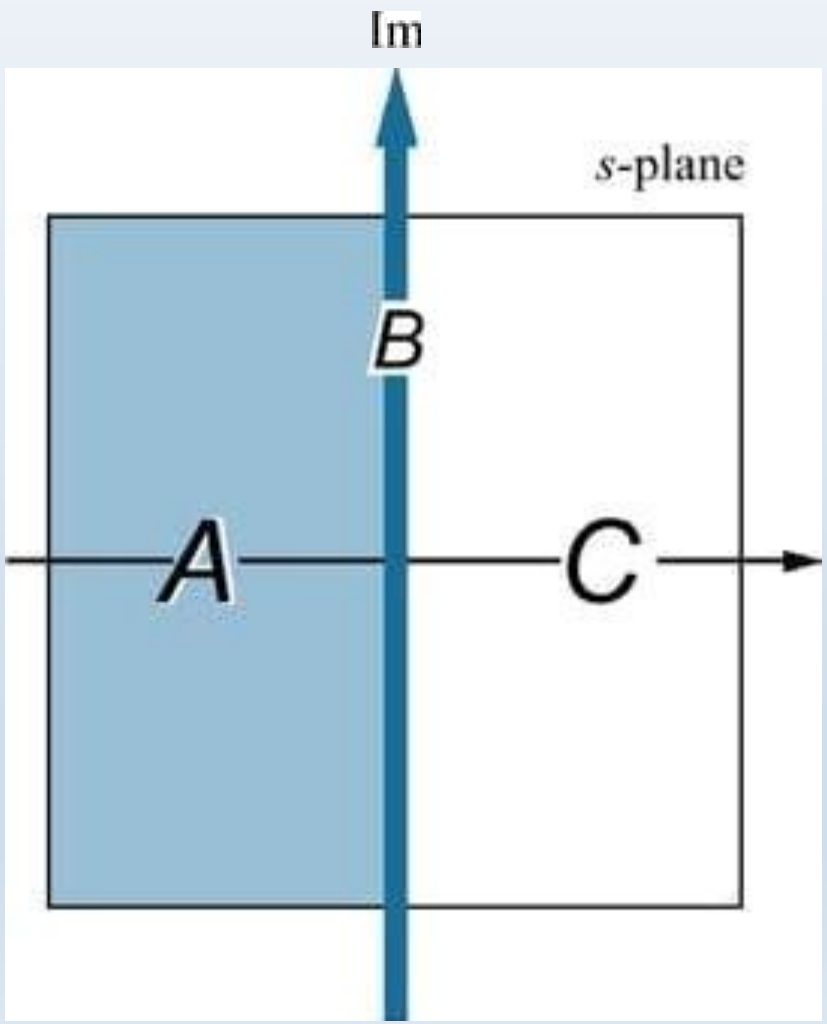
$$z = e^{(\sigma + j\omega)T}$$

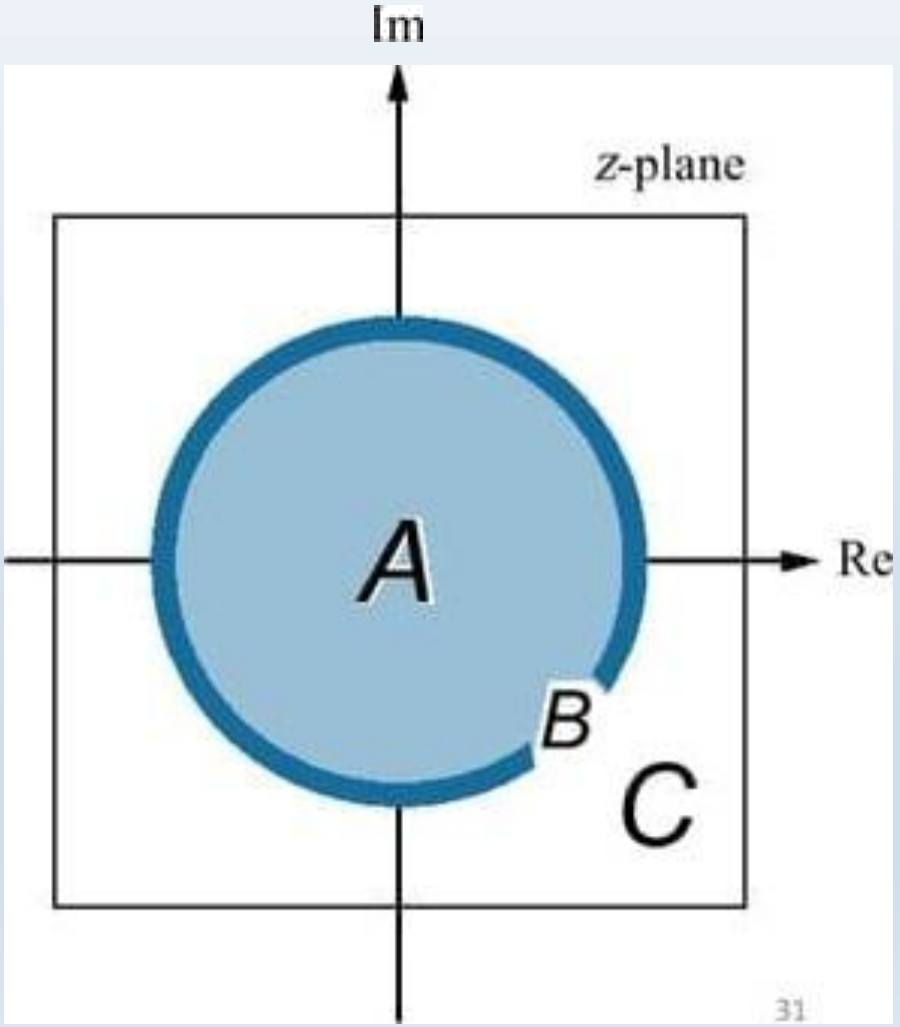
Then z in polar coordinates is given by

$$z = e^{\sigma T} e^{j\omega T}$$

$$|z| = e^{\sigma T} \qquad \angle z = \omega T$$

Mapping regions of the s-plane onto the z-plane

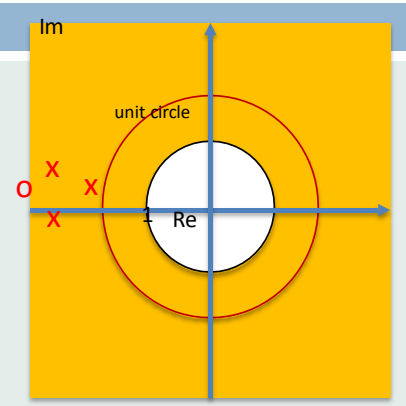




Stable Causal LTI System

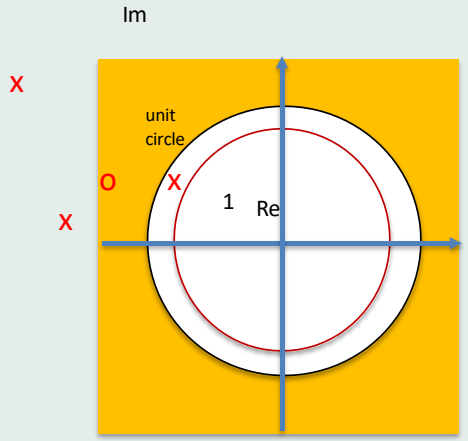
- If the magnitude of each pole is less than one, the poles are less than one unit's distance from the center of the unit circle, and the system is ***stable***. The ROC includes unit circle.

z-plane



Stable System

- If any of the poles of a system lie outside the unit circle, the system is ***unstable***.



Unstable System

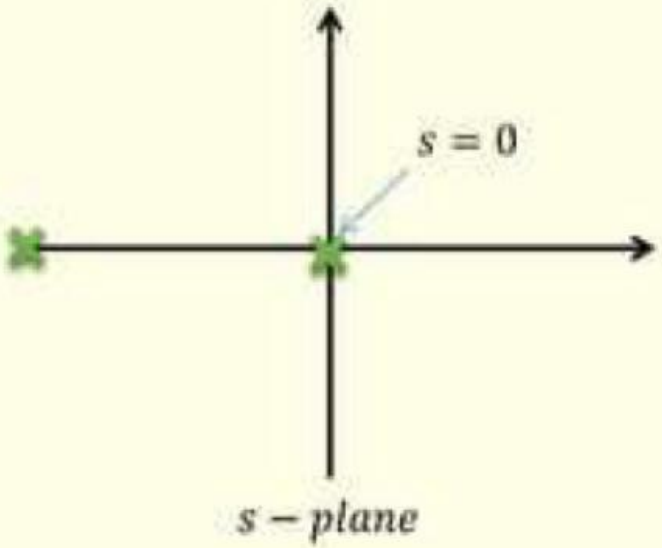
- If the outermost pole lies on the unit circle, the filter is described as being ***marginally stable***

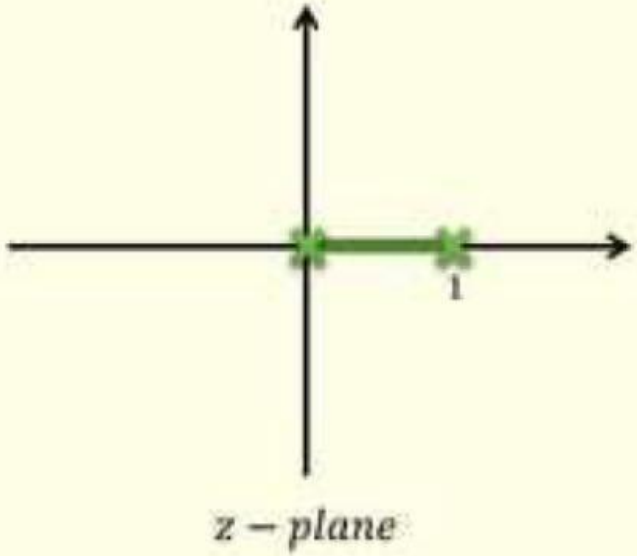
Mapping between s-plane to z-plane

- **Case-1:** Real pole in s-plane ($s = \sigma$)

$$|z| = e^{\sigma T} \qquad \angle z = \omega T$$

- When $s = 0$ $\quad |z| = e^{0T} = 1$ $\quad \angle z = 0T = 0$
- When $s = -\infty$ $\quad |z| = e^{-\infty T} = 1$ $\quad \angle z = 0T = 0$



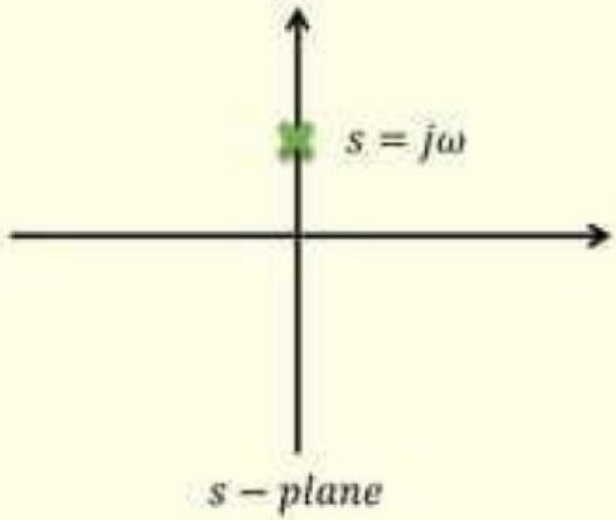


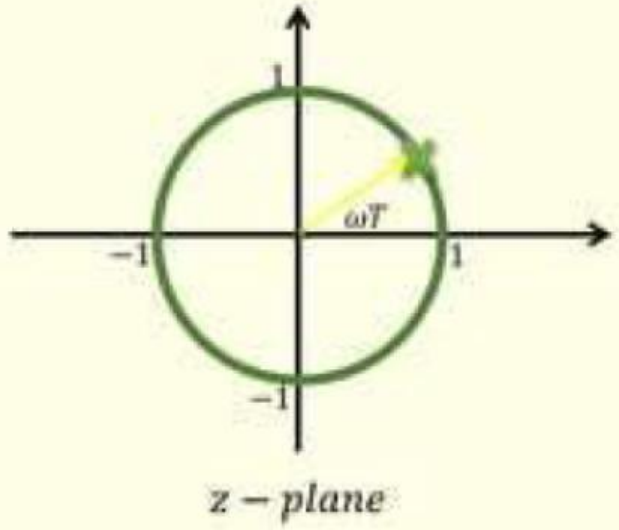
Mapping between s-plane to z-plane

- Case-2: Imaginary pole in s-plane ($s = \pm j\omega$)

$$|z| = e^{\sigma T} \qquad \angle z = \omega T$$

- When $s = j\omega$ $\quad |z| = e^{0T} = 1$ $\quad \angle z = \omega T$

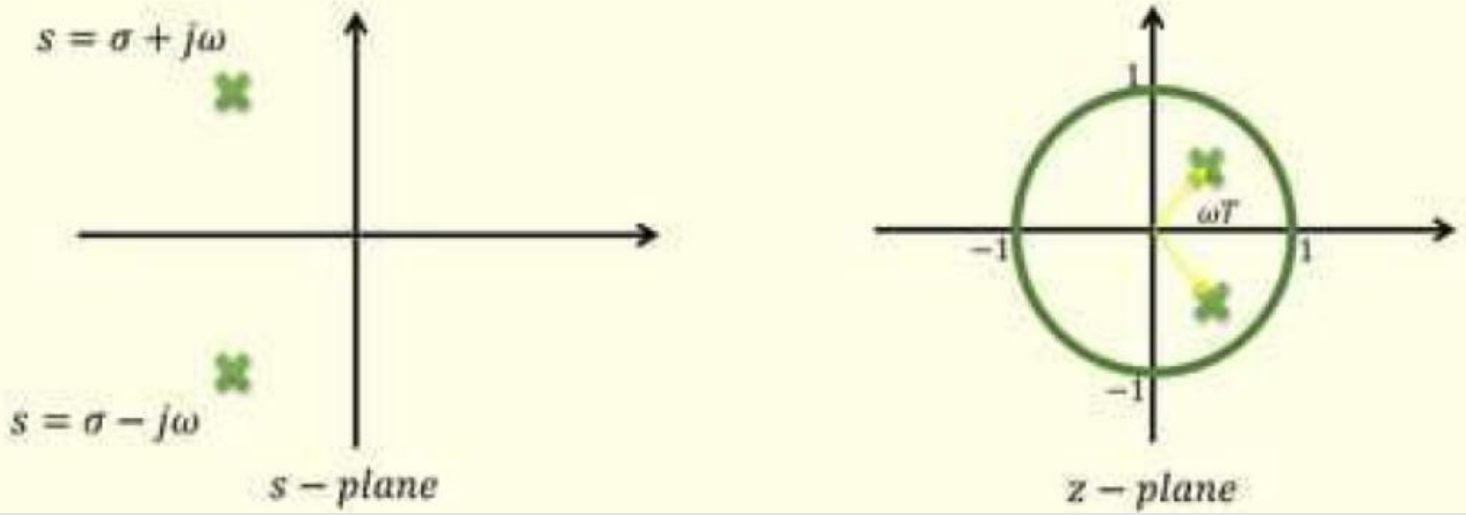




Mapping between s-plane to z-plane

- **Case-3:** Complex pole in s-plane ($s = \sigma \pm j\omega$) $\quad |z| = e^{\sigma T} \qquad \angle z = \omega T$

- When $s = \sigma \pm j\omega \qquad |z| = e^{\sigma T} \qquad \angle z = \omega T$



Example-3

- Map following s-plane poles onto z-plane assume (T=1). Also comment on the nature of step response each case.

1. $s = -3$
2. $s = \pm 4j$
3. $s = \pm \pi j$
4. $s = \pm 2\pi j$
5. $s = -10 \pm 5j$

Solution

Formula:

$$z = e^{sT}$$

Steps:

1. Take the continuous pole $s = a$ (here a is real).
2. Multiply by the sampling period T: compute aT.
3. Exponentiate: $z = e^{aT}$.

Interpretation / stability:

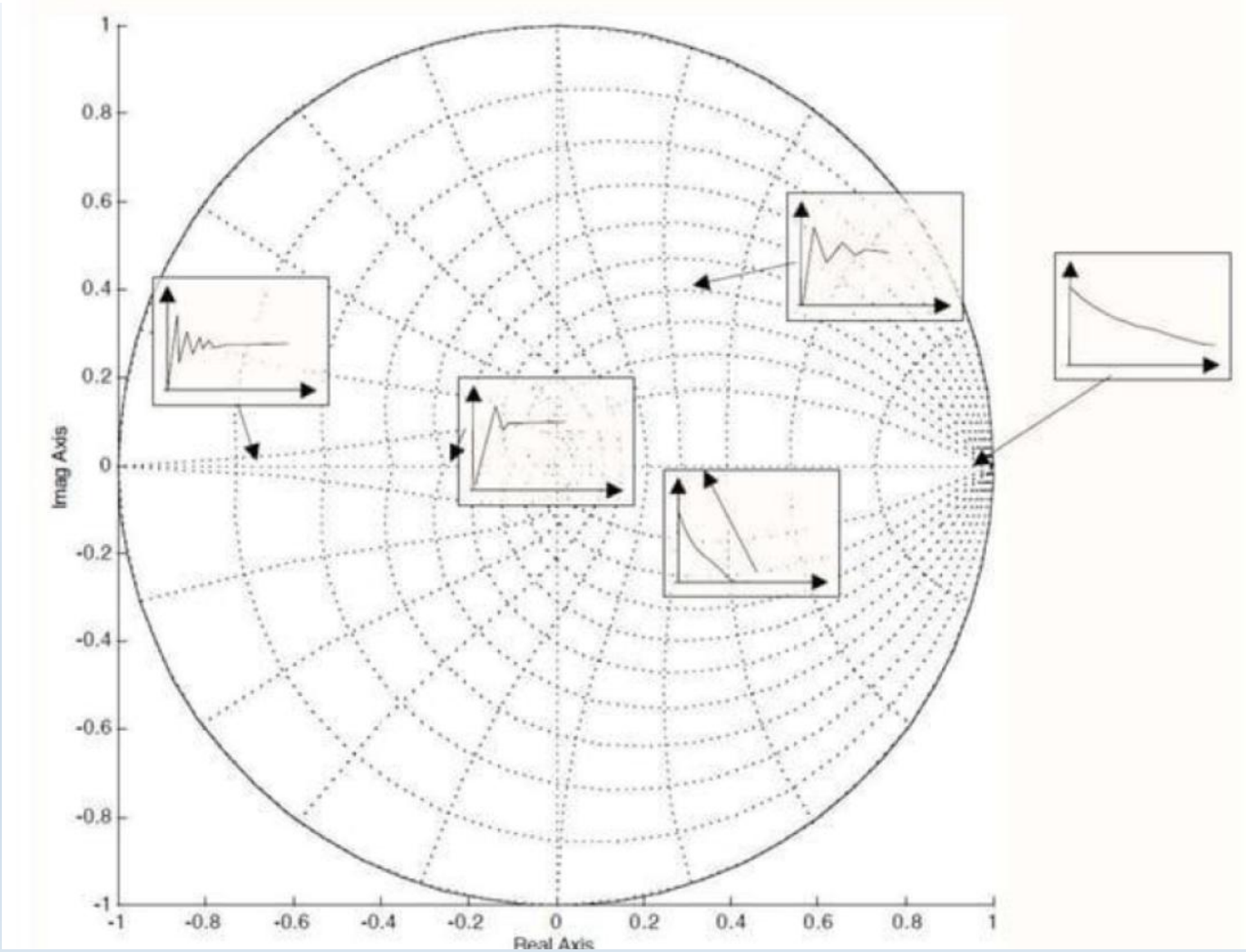
- If $a < 0$ then e^{aT} is between 0 and 1 → mapped **inside unit circle** (discrete stable).
- If $a = 0$ then $z = 1$ → on unit circle (marginal).
- If $a > 0$ then $z > 1$ → outside unit circle (discrete unstable).

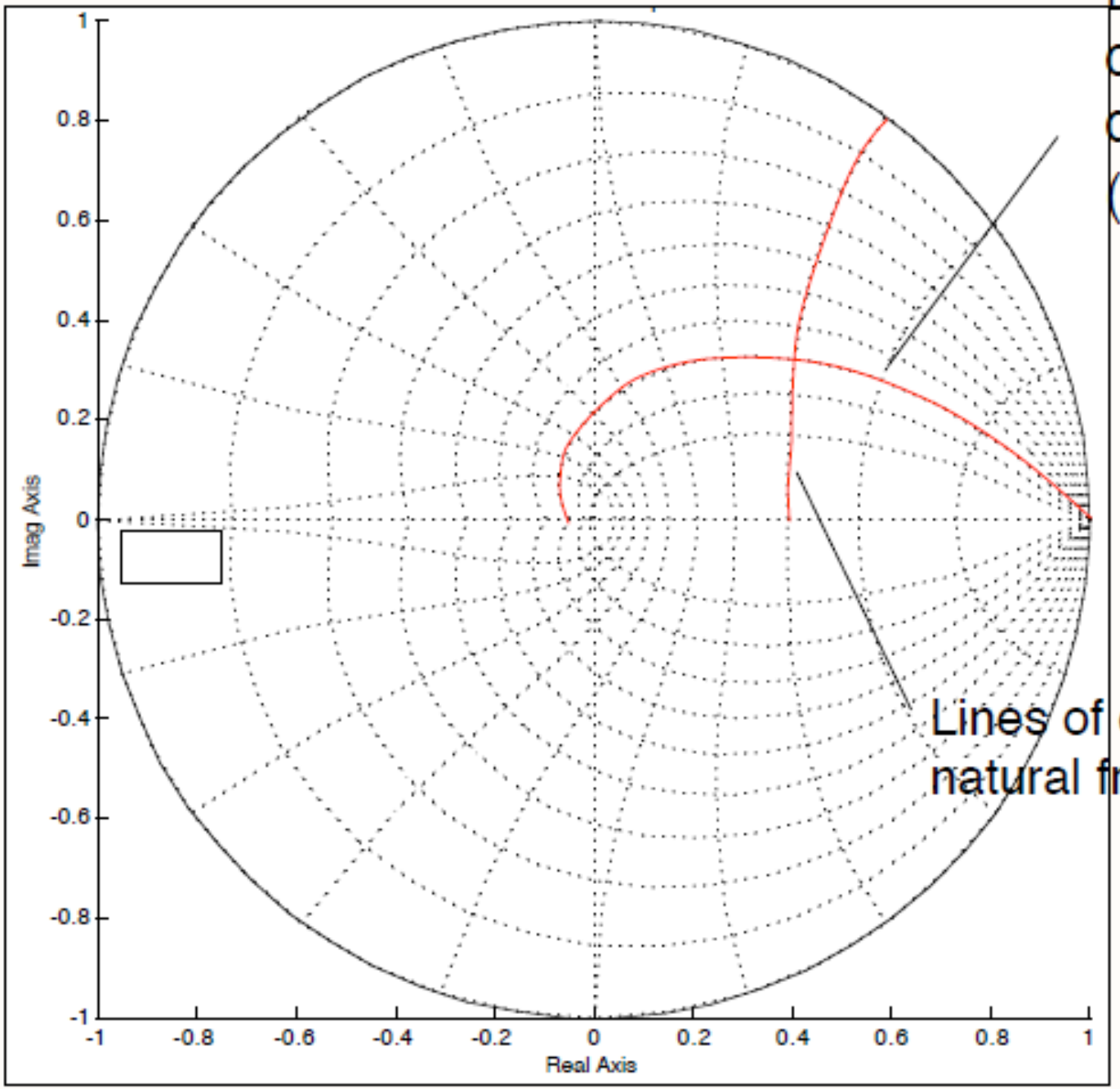
Numeric example (your values): $s = -3,\ T = 1$

1. $aT = -3 \times 1 = -3$.
2. $z = e^{-3}$.
3. $e^{-3} \approx 0.04978706837$.

   So the pole maps to $z \approx 0.04979$ (real, inside unit circle → stable).

Mapping regions of the s-plane onto the z-plane



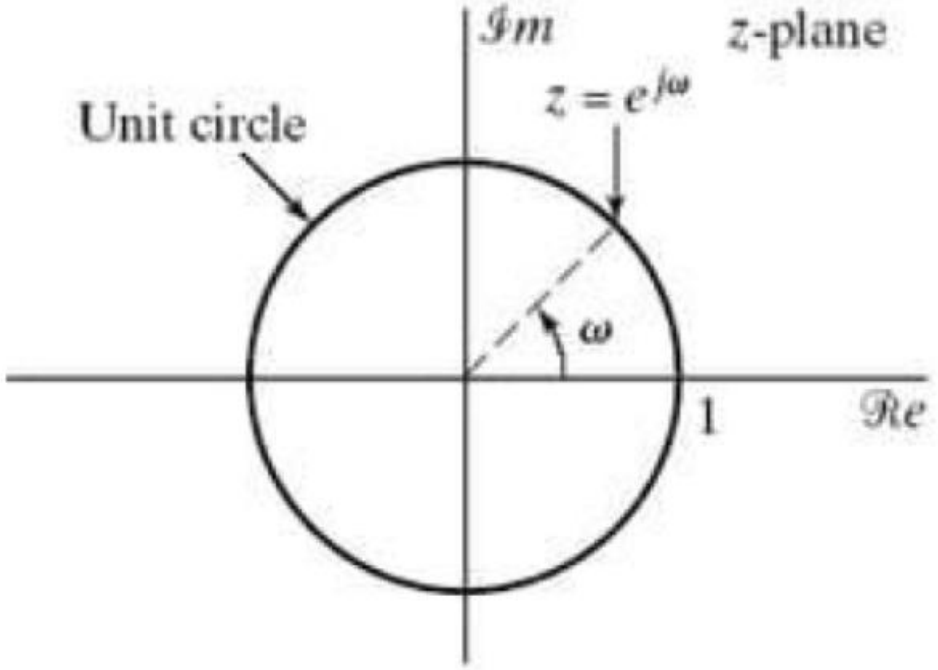


Lines of constant damping ratio (ζ=0.7 shown)

Lines of constant natural frequency

Region of convergence of Z- Transform

- ROC of X(Z) is set for all the values of Z for which X(Z) attains a finite value.



Properties of ROC

- The ROC is a ring, whose center is at origin.
- ROC cannot contain any pole.
- The ROC must be a connected region.
- If ROC of X(Z) includes unit circle then and then only the Fourier transform of DT Sequence x(n) converges.
- For a finite duration sequence x(n); ROC is entire Z-plane except Z=0 and Z=infinite.
- If x(n) is causal then ROC is the exterior part of the circle having radius 'a' and for anti-causal ROC is the interior part of the same.

System Stability Example 1

Find the poles and zeros and stability for the causal discrete-time system whose transfer function is

$$H(z) = \frac{4z^{-1}}{4 - 9z^{-1} + 2z^{-2}}$$

Solution

Eliminating negative exponents yields

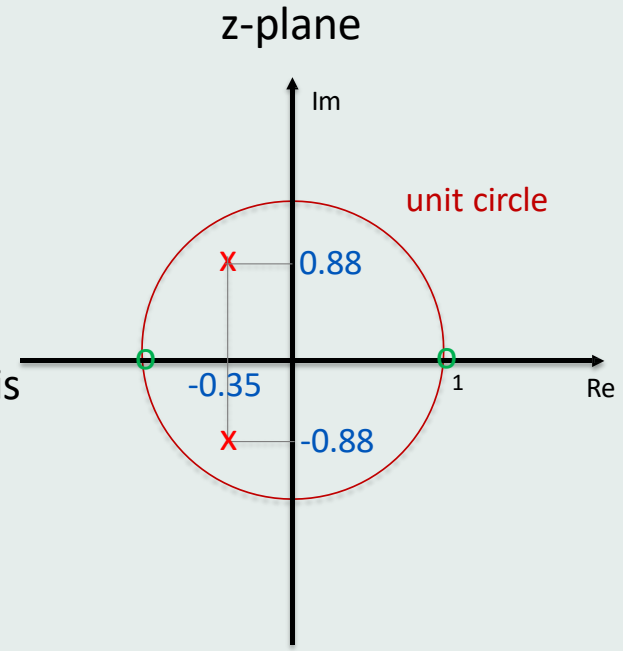
$$H(z) = \frac{4z^{-1}}{4-9z^{-1}+2z^{-2}} = \frac{z^{-1}}{1-2.25z^{-1}+0.5z^{-2}} = \frac{z^{-1}}{(1-0.25z^{1})(1-2z^{-1})} \cdot \frac{z^2}{z^2} = \frac{z}{(z-0.25)(z-2)}$$

- Two Poles at $z = 0.25$ and $z = 2$
- One Zeros at $z = 0$
- As one pole lie outside the unit circle at z = 2, hence the system is ***unstable***.

System Stability Example 2

Find the poles and zeros and stability for the causal discrete-time system whose transfer function is

$$H(z) = \frac{1 - z^{-2}}{1 + 0.7z^{-1} + 0.9z^{-2}}$$



Solution

Two zeros at 0.

Two poles are located at $-0.35 \pm j0.88$

- For these poles the distance from the center of the unit circle is
  - $|z| = \sqrt{(-0.35)^2 + (0.88)^2} = 0.9487 < 1$

As both poles lie inside the unit circle,

Therefore, the system is ***stable***.

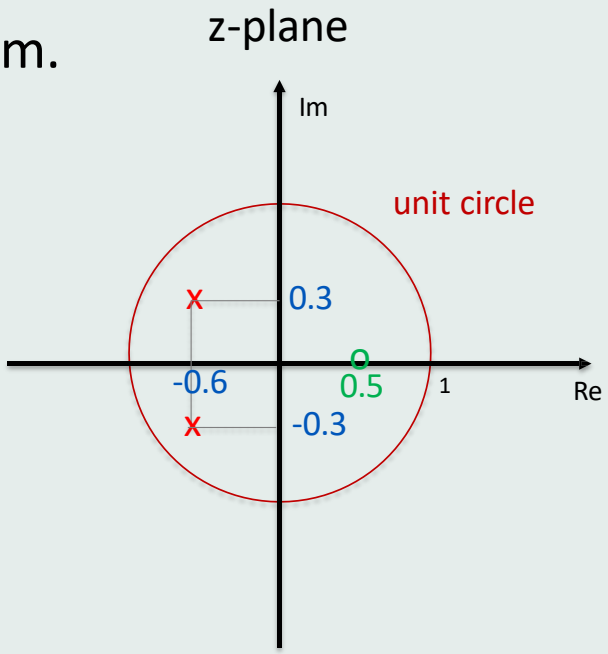
System Stability Example 3

Determine the stability of the following causal system.

$$H(z) = \frac{z^{-1} - 0.5z^{-2}}{1 + 1.2z^{-1} + 0.45z^{-2}}$$

Solution: Eliminating negative exponents yields

$$H(z) = \frac{z^{-1} - 0.5z^{-2}}{1 + 1.2z^{-1} + 0.45z^{-2}} \overset{z^2}{\underset{z^2}{\mathrm{T}}} = \frac{z - 0.5}{z^2 + 1.2z + 0.45}$$



Poles: at $z = -0.6 + j0.3$ and $z = -0.6 - j0.3$

Zero: at $z = 0.5$

As all poles lie inside the unit circle, hence the system is ***stable***.

System Stability Example 4

Find the stability of the filter if the difference equation of the filter is

$$y[n] + 0.8y[n-1] - 0.9y[n-2] = x[n-2]$$

Solution

- Poles are found most easily from the transfer function.

$$H(z) = \frac{z^{-2}}{1+0.8z^{-1}-0.9z^{-2}} \cdot \frac{z^2}{z^2} = \frac{1}{z^2+0.8z-0.9}$$

- The quadratic formula gives the pole locations as

$$z = \frac{-0.8 \pm \sqrt{0.8^2 - 4(1)(-0.9)}}{2(1)} = \frac{-0.8 \pm 2.059}{2} = 0.630 \quad and \quad -1.430$$

- The poles in this case are purely real, without any imaginary component. Clearly the pole at $z = -1.430$ lies outside the unit circle, so the system is **unstable**.

Overview Today's Lecture

- Introduction
- Difference Equations
- Review of Z-Transform
- Inverse Z-transform
- Relations between s-plane and z-plane
- Solution of difference Equations

- Z- Transform of some signals
- With ROC

Signal	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
3. $-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\|z\| < 1$
4. $\delta[n-m]$	z^{-m}	All z except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}}$	$\|z\| > \|\alpha\|$
6. $-\alpha^n u[-n-1]$	$\frac{1}{1-\alpha z^{-1}}$	$\|z\| < \|\alpha\|$
7. $n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\|z\| > \|\alpha\|$
8. $-n\alpha^n u[-n-1]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\|z\| < \|\alpha\|$
9. $[\cos \omega_0 n]u[n]$	$\frac{1-[\cos \omega_0]z^{-1}}{1-[2\cos \omega_0]z^{-1}+z^{-2}}$	$\|z\| > 1$
10. $[\sin \omega_0 n]u[n]$	$\frac{[\sin \omega_0]z^{-1}}{1-[2\cos \omega_0]z^{-1}+z^{-2}}$	$\|z\| > 1$
11. $[r^n \cos \omega_0 n]u[n]$	$\frac{1-[r\cos \omega_0]z^{-1}}{1-[2r\cos \omega_0]z^{-1}+r^2z^{-2}}$	$\|z\| > r$
12. $[r^n \sin \omega_0 n]u[n]$	$\frac{[r\sin \omega_0]z^{-1}}{1-[2r\cos \omega_0]z^{-1}+r^2z^{-2}}$	$\|z\| > r$

Exp, Z-Transform

Obtain the Z-transform of the sequence

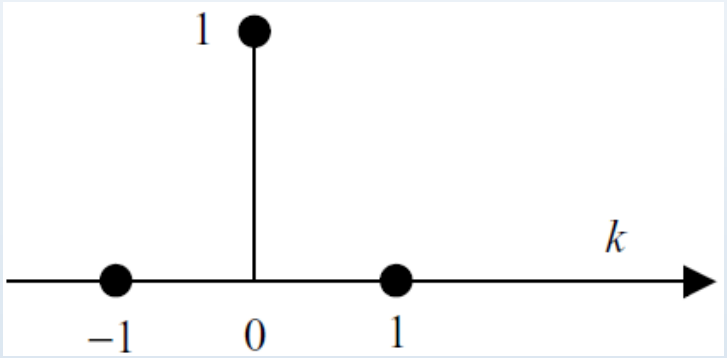
$$x(kT) = \{1, 1, 3, 2, 0, 4, 0, 0, 0, \dots\}$$

Solution:

$$X(z) = 1 + z^{-1} + 3z^{-2} + 2z^{-3} + 4z^{-5}$$

Z-Transform of Unit Impulse Function

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$



- Z-transform of the signal

$$\delta(z) = 1$$

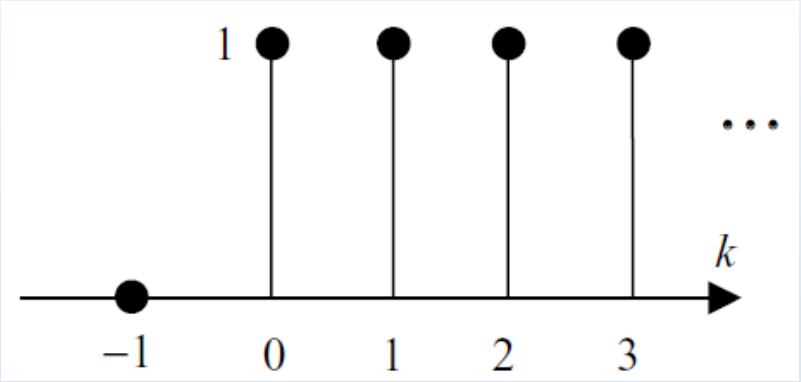
$$\Rightarrow X(z) = \sum_{n=0}^{\infty} \delta(n)\, z^{-n} = 1;\ \text{ROC} \rightarrow \text{All z } i.\ e.\text{entire z} - \text{plane}$$

Or it can also be represented as,

$$\delta(n) \overset{ZT}{\leftrightarrow} 1;\ \text{for all z}$$

Z- transform of unit Step Signal

$$u(k) = \begin{cases} 1, & k \geq 0 \\ 0, & k < 0 \end{cases}$$



$$u(k) = \{1, 1, 1, 1, \ldots\} \qquad k \geq 0$$

$$U(z) = 1 + z^{-1} + z^{-2} + z^{-3} + \cdots + z^{-k} = \sum_{k=0}^{n} z^{-k}$$

$$U(z) = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1} \qquad |z| < 1$$

Z- transform of unit Step Signal

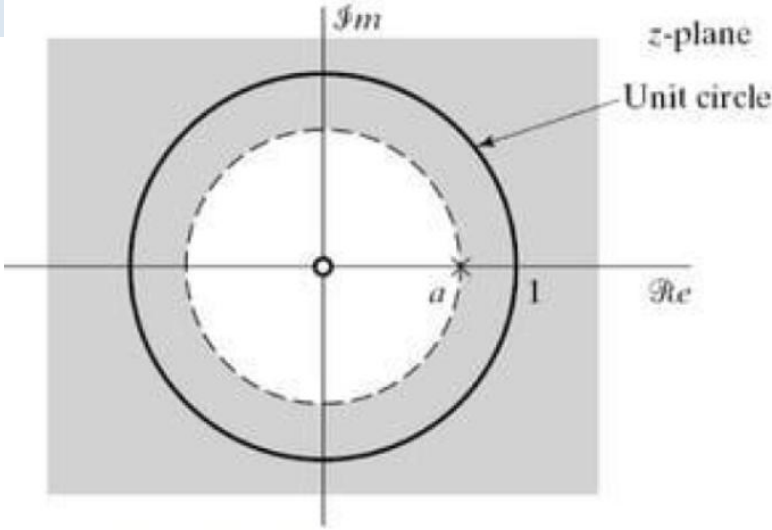
Determine the z-transform, including the ROC in z-plane and a sketch of the pole-zero-plot, for sequence:

$$x[n] = a^n u[n]$$

Solution: $$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} \left(az^{-1}\right)^n = \frac{1}{1-az^{-1}} = \frac{z}{z-a}$$

ROC: $$\left|az^{-1}\right| < 1 \; or \; |z| > |a|$$

$$zero: z = 0 \qquad pole: z = a$$



Gray region: ROC

Z-Transform of Unit Ramp?

- Sampled Ramp

$$x(k) = \begin{cases} k, & k \geq 0 \\ 0, & k < 0 \end{cases}$$

$$Z[x(n)] = X(z) = Z[r(n)]$$

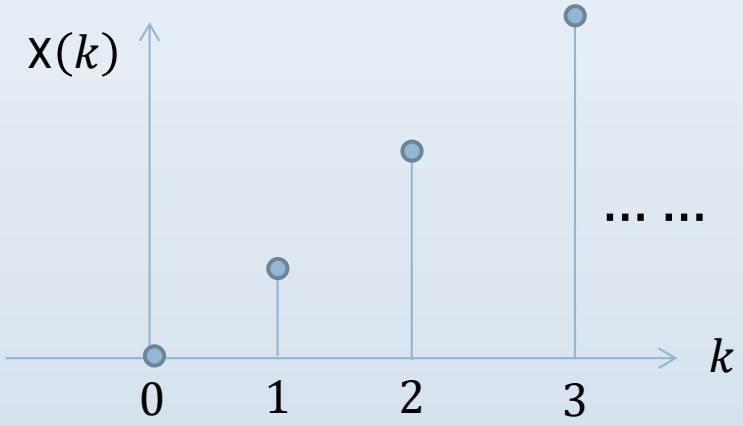
$$\Rightarrow X(z) = \sum_{n=0}^{\infty} r(n)\, z^{-n}$$

$$\Rightarrow X(z) = \sum_{n=0}^{\infty} n z^{-n} = 0 + z^{-1} + 2z^{-2} + 3z^{-3} + 4z^{-4} + \cdots$$

$$\Rightarrow X(z) = z^{-1}\left(1 + 2z^{-1} + 3z^{-2} + 4z^{-3} + \cdots\right)$$

$$\Rightarrow X(z) = z^{-1}\left(1 - z^{-1}\right)^{-2}$$

$$\Rightarrow X(z) = \frac{z^{-1}}{(1 - z^{-1})^2} = \frac{z}{(z-1)^2}$$



This series converges for $|z^{-1}| < 1$. Hence the ROC is $|z| > 1$, i.e., the ROC of the Z-transform of the unit ramp function is the exterior of the unit circle in the z-plane.

Example 1

Find the z-transform of the signal $x[n]$ defined by

$$x[n] = u[n] - (0.5)^n u[n]$$

- Applying the Linear property of the z-transform, we have

$$X(z) = Z\{x[n]\} = Z\{u[n] - (0.5)^n u[n]\}$$

$$= Z\{u[n]\} - Z\{-(0.5)^n u[n]\}$$

$$= \frac{z}{z-1} - \frac{z}{z-0.5}, \quad ROC = |z| > 1$$

Example 2

Find the z-transform of the signal $x[n]$ defined by

$$x[n] = [3(2)^n - 4(3)^n]u[n]$$

- Applying the linearity of the z-transform, we have
  - As we know $Z\{(a)^n u[n]\} = \frac{1}{1 - az^{-1}} \quad ROC: |z| > |a|$
  - Applying the linearity of the z-transform, we have

$$X(z) = 3\frac{1}{1 - 2z^{-1}} - 4\frac{1}{1 - 3z^{-1}} \qquad ROC: |z| > 2 \cap |z| > 3$$

$$= 3\frac{z}{z - 2} - 4\frac{z}{z - 3} \qquad ROC: |z| > 3$$